PUBLIC UNIVERSITY RESEARCH AND ENGINEERING WORK

TI Piccolo PMSM *dynamometer.*

A dual-motor dynamometer built to develop, exercise and validate PMSM control across embedded firmware, CAN communications, real-time testing and measured torque-speed data.

F28069M FIELD-ORIENTED CONTROL QEP CAN REAL-TIME TESTING

01 / CONTEXT

Control at *testbed scale.*

This work was completed in a university laboratory setting. Jonathan's documented contributions span dual-motor control integration, SCI-to-CAN migration, real-time test integration and validation workflows for speed, torque, power and efficiency data.

It is not a private client engagement or an exclusively owned product. The page links to the public repository without reproducing models or large source files.

02 / ARCHITECTURE

Motors, controls *and evidence.*

Technical scope

The public material documents TI C2000 F28069M control, field-oriented control architecture, QEP encoder integration, CAN communications, Speedgoat/MATLAB real-time testing where documented, efficiency mapping and validation data processing.

SYSTEM

Dual-motor PMSM dynamometer integration.

METHODS

Torque-speed, power and efficiency validation workflows.

TESTBED ARCHITECTURE PLACEHOLDER

DUAL-MOTOR PMSM → F28069M → CAN → REAL-TIME TEST

Authorized dynamometer architecture diagram.

Source: public repository architecture documentation

SIGNAL MAP PLACEHOLDER

CAN AND QEP CHANNELS

CAN signal map or controller/testbed photograph.

MEASURED DATA PLACEHOLDER

SPEED / TORQUE / POWER

Measured plot with operating point and instrument context.

EFFICIENCY MAP PLACEHOLDER

VALIDATION RESULT

Efficiency map when source and publication authorization are confirmed.

03 / EVIDENCE

Measured work *with boundaries.*

Repository documentation, hardware integration notes, MATLAB/Python processing and measured plots support the scope described here. Complete instructional exercises, raw Simulink models, sponsor material and credentials remain outside this site.

VIEW REPOSITORY ↗

RELATED SERVICE: TEST AUTOMATION →

START A TECHNICAL DISCUSSION

Bring the *technical question.*

Describe the system, its current stage and the evidence available. Secure file exchange can follow the initial discussion.

START A TECHNICAL DISCUSSION →

Or write directly: info@herdersystemen.com

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